

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.6

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LS 1.1.6- DRUG EXCRETION

DRUG EXCRETION

LS 1.1.6: LEARNING OBJECTIVES:

- Define the term “Excretion” as it is used in pharmacokinetics
- Describe the processes of drug excretion
- Describe the processes involved in renal excretion of drugs
- Explain enterohepatic circulation and its therapeutic significance

DRUG EXCRETION

INTRODUCTION:

Excretion: This is the process by which a drug or metabolite is eliminated from the body without further chemical change

Kidneys are the major organs of excretion

Lungs are the main route of elimination of volatile anaesthetics. Other than this they play a trivial role in drug elimination.

DRUG EXCRETION

INTRODUCTION ... CONT'D:

Biliary excretion (bile and feces) is important for some drugs. These drugs may be reabsorbed when passing through the intestines from the liver (enterohepatic recirculation).

Intestines, sweat, saliva and breast milk constitute minor routes of drug excretion

RENAL EXCRETION

Renal excretion of drugs occurs chiefly by the following three processes:

1. Glomerular filtration
2. Passive tubular re-absorption
3. Active tubular secretion

Total renal excretion = Excretion by ***filtration*** plus
Excretion by ***secretion*** minus Retention by ***re-absorption***

GLOMERULAR FILTRATION

- 20% of renal blood flow is filtered through the glomerulus therefore 20% of drug in plasma is delivered to the glomerulus
- Drug molecules of molecular weight below 20,000 daltons diffuse into the glomerular filtrate (most drugs fall in this category)
- Substances with molecular weight $> 50,000$ are excluded from the glomerular filtrate
- Drug that is bound to plasma protein is not filtered

ACTIVE TUBULAR SECRETION

- At least 80% of drug delivered to the kidney is brought to the renal tubules
- Cells of the proximal renal tubule actively transfer charged molecules from the plasma to the tubular fluid
- A drug can be secreted into the tubular lumen even when it is bound to plasma protein

ACTIVE TUBULAR SECRETION CONT'D

- Drug molecules are actively secreted into the tubular lumen by two independent and relatively nonselective carrier systems, one for acidic drugs and the other for basic drugs
- Many of the drugs that are excreted by the kidney share the same transport system and therefore competition can occur between them, leading to drug interactions. For example probenecid reduces the excretion of penicillin.

ACTIVE TUBULAR SECRETION CONT'D

Examples of drugs that are actively secreted into the proximal renal tubule:

- Acids: frusemide, methotrexate, indomethacin, salicylic acid, penicillins
- Bases: amiloride, pethidine, quinine

PASSIVE TUBULAR REABSORPTION

- After the drug molecule or its metabolite is filtered and/or actively secreted into the tubular lumen, the concentration of drugs in the renal tubule is more concentrated than in plasma
- Depending on their lipid solubility and pH of the tubular fluid, some molecules passively diffuse back into plasma i.e. they are reabsorbed
- Lipid soluble molecules are easily reabsorbed by passive diffusion and are therefore not efficiently excreted in the urine

PASSIVE TUBULAR REABSORPTION.... CONT'D

- Many drugs being weak acids or weak bases change their ionization status with pH. This can affect their renal excretion
- The more ionized a molecule is, the less it would be passively reabsorbed
- Basic drugs will ionize in acidic urine while acidic drugs ionize in alkaline urine; with poor reabsorption in both cases, thus favouring renal excretion
- Manipulation of urinary pH can therefore be done to facilitate the excretion of certain drugs through making them more ionized e.g. sodium bicarbonate is given to alkalinize the urine to treat aspirin overdose (aspirin is an acidic drug)

BILIARY EXCRETION AND ENTEROHEPATIC CIRCULATION

- Various substances including drugs are transferred from plasma to bile by liver cells by means of transport systems similar to those of the renal tubule
- The bile delivers the drug to the intestine from where it is eliminated in faeces
- Various hydrophilic drug conjugates (particularly glucuronides) are concentrated in bile and delivered to the intestine

BILIARY EXCRETION AND ENTEROHEPATIC CIRCULATION

- The glucuronides are too polar to be reabsorbed and therefore remain in the gut where they usually hydrolysed (by enzymes and bacteria), releasing active drug once more
- Free drug can then be reabsorbed and the cycle repeated (enterohepatic circulation). This creates a reservoir of drug and prolongs drug action.
- Examples of drugs that undergo enterohepatic circulation are morphine, sulindac and ethinyloestradiol

LEARNING OUTCOMES

Students must be able to comprehend:

- The definitions of “Excretion”
- The processes of drug excretion
- The processes involved in renal excretion of drugs
- The process of enterohepatic circulation and its therapeutic significance

QUIZ

1. Half life ($t_{1/2}$) doesn't depend on:

- a) Biotransformation b) Time of drug absorption c) Concentration of a drug in plasma d) Rate of drug elimination

2. Elimination is expressed as follows:

- a) Rate of renal tubular reabsorption b) Clearance speed of some volume of blood from substance c) Time required to decrease the amount of drug in plasma by one-half d) Clearance of an organism from a xenobiotic

3. The most rapid eliminated drugs are those with high glomerular filtration rate and actively secreted but aren't passively reabsorbed:

- a) True b) False

END

Thanks for listening